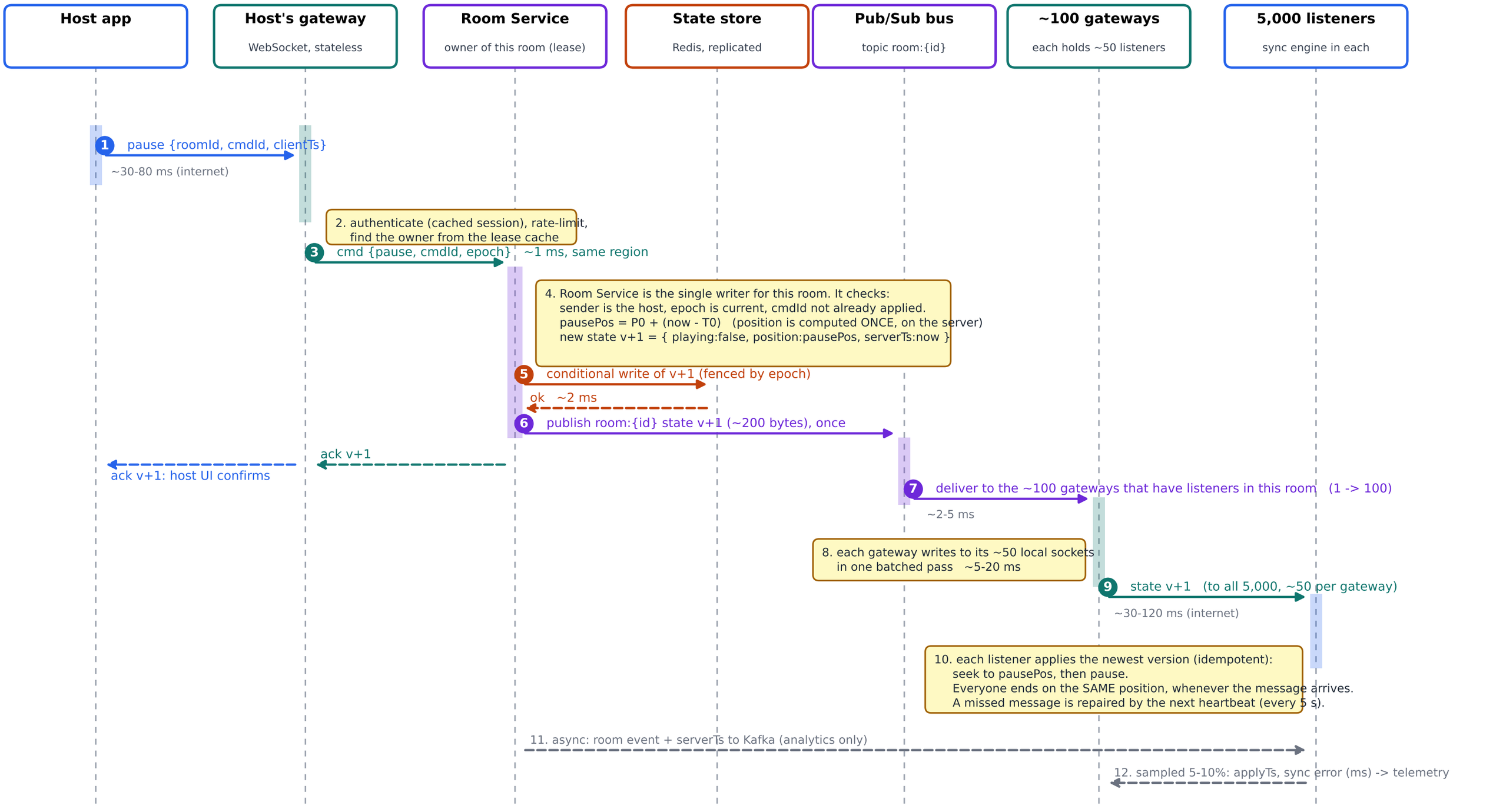
Sequence: host presses pause in a room of 5,000 listeners



If the owner node dies mid-command
- before step 5: the host retries with the same cmdId.
  A new owner takes the lease within ~5 s (epoch +1);
  duplicates are ignored.
- after step 5: nothing is lost. State v+1 is in Redis and
  listeners keep extrapolating the last anchor, so the
  audio never stops.

A stale owner is blocked: its writes carry an old epoch.

Timing (same-region path; the internet legs dominate)
host -> gateway 30-80 ms | gateway -> room 1 ms | persist 2 ms | publish 2 ms | bus -> gateways 2-5 ms | socket write 5-20 ms
| gateway -> listener 30-120 ms. A listener hears the pause about 70-230 ms after the tap.
What decides how far apart listeners are is only the last leg, not the server. In the simulation (5,000 listeners) their stop times
differ by p95 124 ms, and because the message carries the position, they all land on the same spot.
Fan-out cost is flat: the Room Service sends ONE message per event, whether the room has 2 listeners or 5,000.